



Figure 2: Visualization of the located regions at each iteration. The attentional regions can well locate most semantic objects in the images.

The comparison results are depicted in Table 5. It shows that using multiple regions at each iteration leads to better classification performance.

Method	mAP (%)
multiple regions	91.3
single region	90.9

Table 5: Comparison of mAP in % of our model that extract multiple and single regions at each iteration on the VOC07 dataset. The results are evaluated using ten-view with the scale of 512×512 .

Analysis of increasing the recursive iteration In this part, we explore the effect of using different recursive iterations T . To this, we train our model with different iterations, i.e., $T = 1, 5, 10$, and report the experimental results in Table 6. When the iteration increases from 1 to 5, the performance has a notable improvement since the located regions may cover more discriminative objects. However, when further increasing the iteration number, the performance does not improve. One possible reason is that when the iteration is greater than 5, the agent has almost mined out all discriminative regions, and locating more regions make litter sense or even bring noise and redundant computation. Thus, in our experiments, the iteration number is set as 5 to better balance the efficiency and effectiveness.

T	mAP (%)
1	90.9
5	91.3
10	91.3

Table 6: Comparison of mAP in % of our model using different recursive iterations on the VOC07 dataset. The results are evaluated using ten-view with the scale of 512×512 .

Efficiency analysis

Efficiency is another important metric for the real-world systems. In this part, we analyze the execution time of our model and the previous state-of-the-art methods. We test our model on a desktop with a single NVIDIA GeForce GTX TITAN-X GPU. It takes about 150ms for ten-view evaluation for scale 512 and about 200 ms for scale 640. Thus, the execution time

of our method is about 350ms per image. However, recent proposal-based methods, e.g., HCP (?) and FeV+LV (?), need to compute the proposals and repeat processing hundreds of proposals using the deep CNNs, rendering them extremely inefficient. As shown in (?), these methods may take about 10s to process an image on a similar GPU environment, about $30\times$ slower than ours.

Conclusion

In this paper, we propose a recurrent attention reinforcement learning framework that is capable of automatically locating the attentional and informative regions regarding classification, and predicts the label scores over all attentional regions. We formulate the region localization process as a sequential decision-making problem and resort to reinforcement learning technique to optimize the proposed framework with merely image-level labels in an end-to-end manner. Compared to the previous proposal-based methods, our method can better explore the interaction between semantic labels and attentional regions, while explicitly capturing the contextual dependencies among these regions. Extensive experimental results and evaluations on two large-scale and challenging benchmarks, i.e., Pascal VOC and MicroSoft COCO, well demonstrate the superiority of our proposed method on both accuracy and efficiency.

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